MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8901 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION:

Reflex, integral cavity, mechanically tunable, 23.6 to 24.4 GHz, waveguide output, air cooled.

ABSOLUTE RATINGS:

Parameter:	Ef	If	Ers	Irs	Er	F	T
Unit:	V	A	Vdc	mAdc	Vdc	GHz	°C
Maximum:	7.0	1.32	400	40	-500	24.4	200
Minimum:	5.7	----	---	----	-20	23.6	----

PHYSICAL CHARACTERISTICS:

Dimensions: See Figure 1 Weight: 4.5 oz (maximum)
Unit: Cathode - Unipotential

TEST CONDITIONS: - See Note 1

Parameter:	Ef	Ers	Er	F	T	VSWR
Unit:	V	Vdc	Vdc	GHz	°C	----
	6.3	375	-270 to -350	24.0	160	1.1:1
	See note 12		Adjust/Max Po		See note 2	

REQUIREMENT OR TEST:

First Article Testing (Appendix F) - Required

1/ Replaces T.I. P/N 506751-1 and VA98M

8901

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification					
1042	Shock	Test Condition C No Voltages				
----	Post Shock Tests:	Change in frequency Change in power output Reflector modulation	ΔF ΔPo/Po ΔF/ΔEr	--- --- ---	5 5 2	MHz % MHz/V
----	Long term stability	See Note 14	ΔF ΔPo/Po	--- ---	5 5	MHz %
	Low pressure operation	55 mmHg; t = 10 sec; F = 24.0 GHz; See Note 15	ΔF	---	3	MHz
1027	Temperature cycling	TA = -62°C to 95°C 1 cycle; F = 24.0 GHz See Notes 6 and 13	ΔPo1/Po ΔPo2/Po	--- ---	10 10	% %
...	Quality conformance inspection, part I	See Note 3				
----	High frequency vibration	G = 10; F = 60 Hz; T = 120; See Note 9	Ir	---	5	μAdc
4214	Cathode emission	Ef = 5.7 Vac; See Note 10	$\frac{\Delta Irs}{Irs}$	---	15	%
1301	Heater current	Ef = 6.3 Vac	If	0.3	1.32	Aac
4280	Electronic tuning range	F = 23.6 and 24.4 GHz Er/50% max Po	ΔF	50	---	MHz
4250	Power output	F = 23.6, 24.0 and 24.4 GHz	Po	100	---	mW
1256	Resonator current	Er = -100 Vdc; See Note 10	Irs	18	40	mAdc
4213	Reflector voltage	F = 23.6 and 24.4 GHz	Er	-270	-350	Vdc
4231		See Note 11	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2	See Note 5				
4027	Temperature coefficient	TA = -62°C to 95°C See Note 7	ΔF/ΔT	—	±200	kHz/°C
1031	Variable frequency vibration	F = 5-500-5 Hz; See Note 4	—	—	—	—
—	Post Vibration Tests	Frequency modulation	ΔF	—	2	MHz (P-P)
		Change of mode width	ΔBW	—	5	%
		Change in frequency	ΔF	—	5	MHz
		Change in power output	ΔPo/Po	—	5	%
—	Resonator modulation sensitivity		ΔF/ΔErs	—	2	MHz/V
	Quality conformance inspection, part 3					
	Life test	Group D, See Note 12	t	1000	—	hours
	Life Test End Point					
4250	Power output		Po	90	—	mW

NOTES:

1. All oscillation tests (except vibration test) shall be made with the tube rigidly connected to a UG-595/U flange. The load VSWR for the tube shall be less than 1.1:1. Envelope temperature should not exceed 160°C.
2. While forced air cooling is normally required, the tube may be operated with conductive cooling through the associated waveguide, provided the heat sink is such that the envelope temperature of the tube is held to +160°C or less.
3. All quality conformance inspection, part 1, tests shall be performed after the Holding Period.
4. The tube shall be fastened rigidly to the vibration platform and vibrated with simply harmonic motion over the following frequency ranges:

   a. Frequency Range = 5 - 45 Hz
      Displacement = 0.1" (Min)
      Sweep Time = 3 Minutes (Min); 3.5 Minutes (Max)

   b. Frequency Range = 45 - 500 Hz
      Acceleration = 10G (Min)
      Sweep Time = 3.5 Minutes (Min); 4 Minutes (Max)

   Each tube shall be vibrated for 2 hours in each of the three positions X, Y, and Z.

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NOTES: - Continued

5. This test shall be performed on one tube per month during production.

6. The tube shall meet Method 4250 requirements after this test.

7. Connect a heat sink to the tube as shown in Figure 2. The output flange of the tube shall be connected to a standard mating flange and waveguide, with a teflon spacer between the output flange of the tube and the mating flange. The tube shall be cooled with air at the rate of 32 ft^3/min when the ambient temperature is above +10°C. The blower shall be turned off when ambient temperature is below +10°C. The temperature coefficient shall be measured by the following procedure:

   a. Stabilize the tube at 25°C ambient with the reflector voltage adjusted for maximum power output.

   b. Lower the ambient temperature to -62°C and hold until the tube is stabilized at this temperature. Apply filament voltage only for 4 minutes, then apply all voltages and measure the frequency, F_1. Do not adjust the reflector voltage. Then, measure the body temperature, T_1, at the point indicated on Figure 1. F_1 shall be measured within 10 seconds after application of the high voltage.

   c. Turn off all voltages and increase the ambient temperature to +95°C. After the tube is stabilized at this temperature, turn on the blower and measure F_2. Do not adjust the reflector voltage. Measure the body temperature, T_2, at the point indicated on Figure 1. F_2 shall be measured within 10 seconds after the application of high voltage.

   d. Temperature coefficient shall be computed from the following relationship:

$$\Delta F/\Delta T = (F_2-F_1)/(T_2-T_1)\text{kHz}/^\circ\text{C}$$

8. The tube shall be tested for full power output at the three (3) frequencies specified, however, the tube must be capable of oscillation at any frequency between 23.6 and 24.4 GHz.

9. The reflector current shall be recorded with a "Brush Model BL-202" Recorder or equivalent. There shall be no reflector current bursts greater than the value specified. Set Ers to 300V, Er/max Irs, and set frequency to any point within the band. The tube need not be oscillating.

10. The tube shall not be oscillating during the test.

11. Superimposing sufficient 60 cycles ac voltage on the direct reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope. There shall be no discontinuity in the reflector voltage mode between the lower frequency 20% power point and the upper frequency 50% power point for any phase of the load VSWR of 1.1:1.

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NOTES: - Continued

12. The heater voltage shall be applied for two minutes prior to application of the resonator voltage.

13. The tube shall operate over an ambient temperature range of -62°C to +95°C with a maximum 32 ft^3/min of forced air cooling the tube. Forced air cooling may be removed below 10°C.

    a. Connect tube to standard flange, waveguide, and heat sink as shown in Figure 2. With blower operating, all voltages applied, stabilize the tube at 25°C ambient temperature and adjust the reflector for maximum power output. Measure power output - this is Po.

    b. Turn all voltages and blower off and lower ambient temperatures to -62°C. Hold at this temperature for 2 hours.

    c. Increase ambient temperature to -54°C and hold for 2 hours. (All voltages and blower off).

    d. Apply heater voltage only for 4 minutes, than apply all voltages and read the power output within 10 seconds - this is Po1. Do not adjust the reflector or turn the blower on.

    e. Turn off all voltages and increase ambient temperature to +95°C.

    f. Turn on blower, apply heater voltage for a period of 4 minutes, then apply all voltages and within 10 seconds read the power output - this is Po2. Do not adjust the reflector.

    Po2 must be within ±10% of Po
    Po1 must be within ±10% of Po.

14. With the tube operating from constant input voltages, the power output and frequency shall not vary greater than the values shown, over a period of 50 hours.

15. The frequency shall be stabilized at a pressure of 55 mmHg. Adjust the reflector voltage for maximum power output. The pressure shall then be increased to 760 mmHg in the time specified and the frequency again read. The resulting frequency change shall not exceed the limit specified. Do not readjust the reflector voltage.

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Air Force - 80

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User activity:
Air Force - 11

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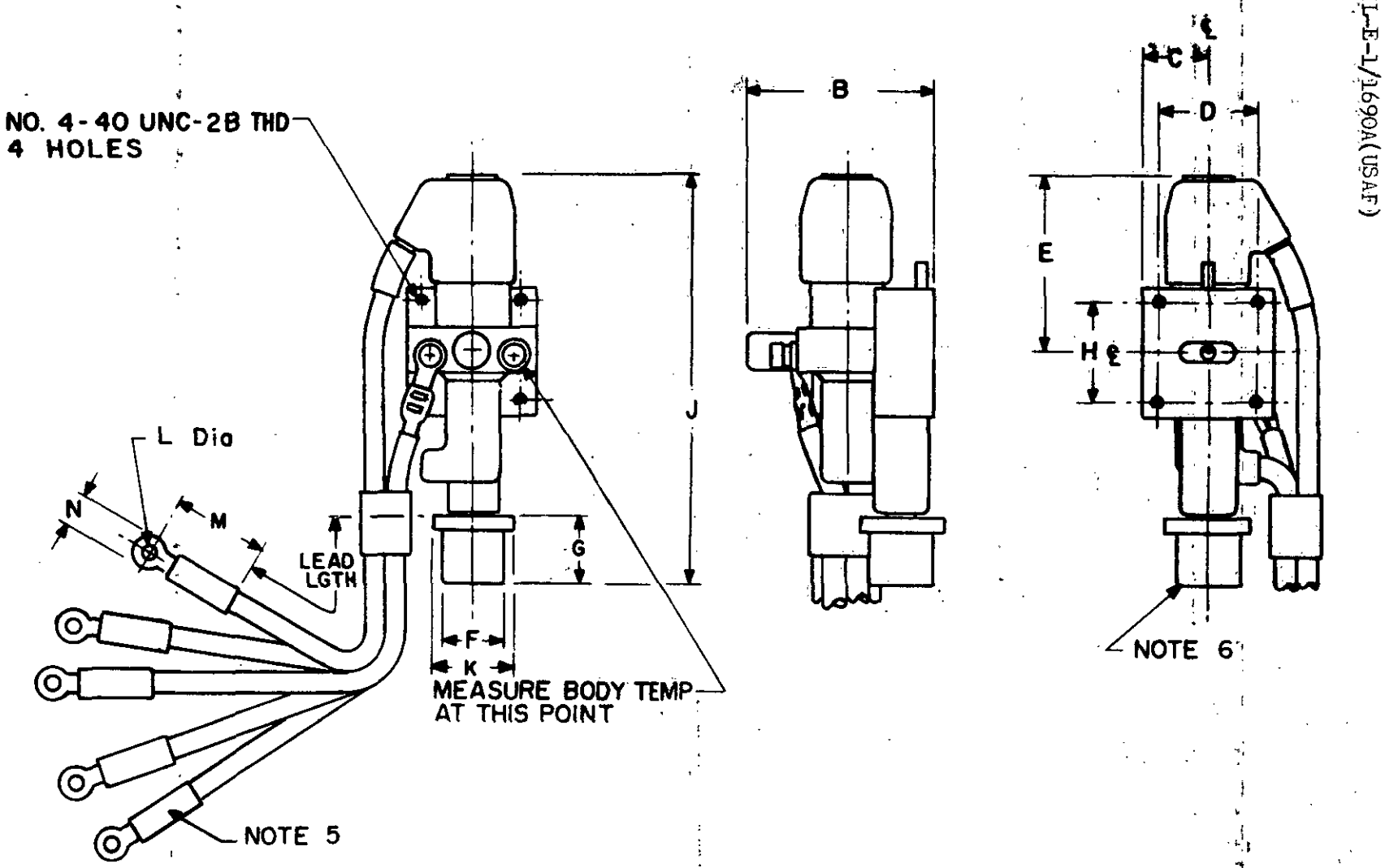


FIGURE 1 Outline drawing of electron tube type 8901

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DIMENSIONS									
LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
				60.33	● H	.665	.675	16.89	17.15
B		1.375		34.93	● J		2.937		74.60
C	.428	.448	10.87	11.38	K		.645		16.38
● D	.635	.645	16.13	16.38	L	.120	.124	3.05	3.15
E		1.328		33.73	M		.700		17.78
F	.400	.450	10.16	11.43	N		.250		6.35
G	.450	.510	11.43	12.95					

● *DIMENSIONS, QUALITY CONFORMANCE INSPECTION, PART 1*
ALL OTHER DIMENSIONS, QUALITY CONFORMANCE INSPECTION, PART 2

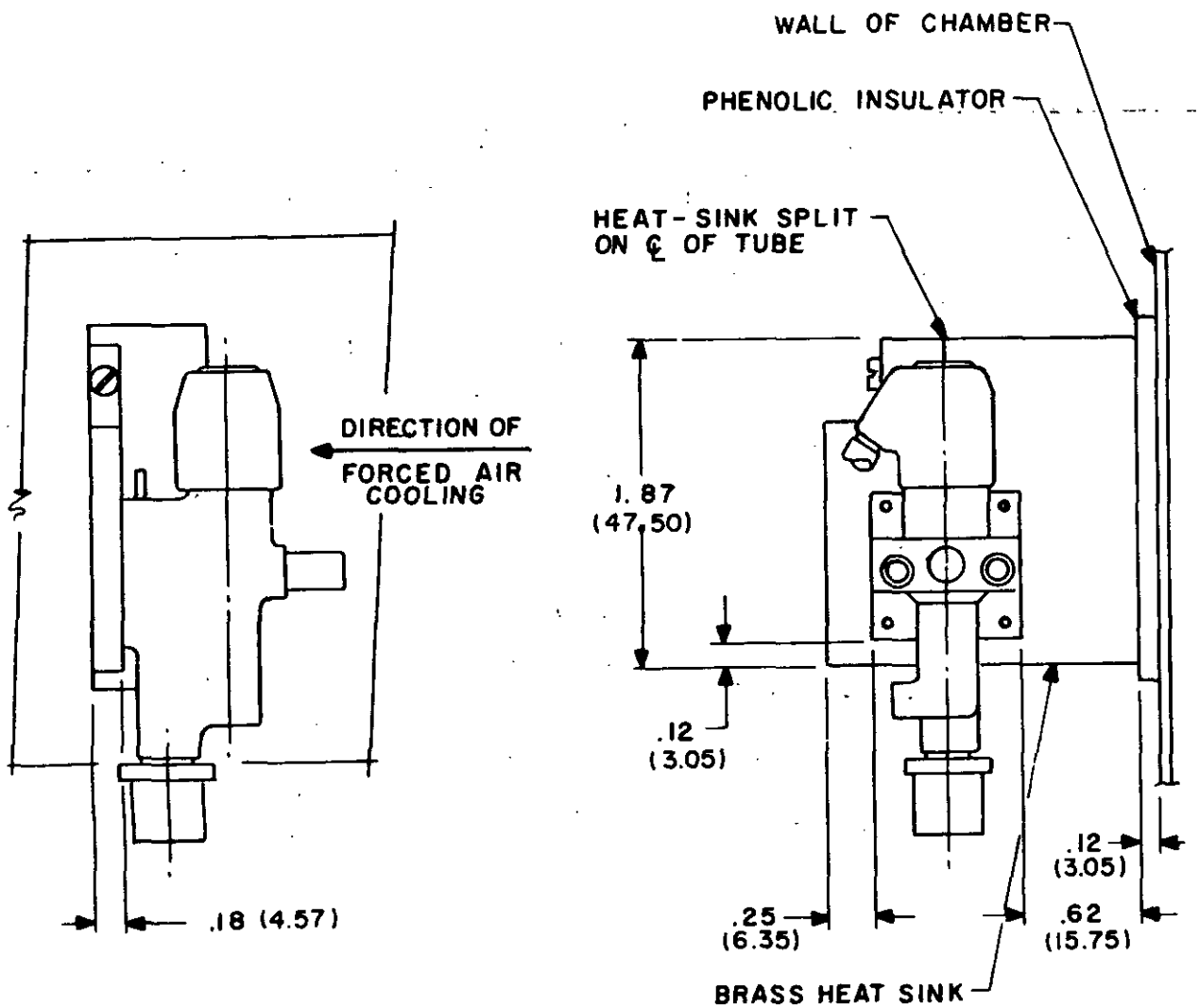
BASE CONNECTORS				
	Term NO.	Lead Color	Lead Length	Element
	1	GRAY	4.00 (101.6 MM)	REFLECTOR
	2	BROWN	4.25 (107.95 MM)	BODY
▷	3	WHITE	4.50 (114.30 MM)	HEATER
	4	YELLOW	4.75 (120.65 MM)	HEATER
▷	5	GREEN	5.00 (127.0 MM)	CATHODE

▷ INTERNALLY CONNECTED

NOTE:
1) METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION AND ARE BASED ON 1 INCH = 25.4 MM
2) OUTPUT CONNECTOR:- FLANGE MATES WITH UG-595/U
3) LEAD LENGTH MEASURED FROM THE TOP OF THE REPELLER CAP
4) LEAD TERMINATIONS:- AMP INC PART NO. 31094 OR EQUAL
5) MARK LEAD NOS. PER BASE CONNECTOR CHART ON SLEEVE
6) PLASTIC ROD, POLYAMIDE, RIGID TYPE I PER MIL-P-17091
7) MOUNTING POSITION:- ANY
8) LEADS:- SILVERED SOFT COPPER CONDUCTOR NO. 22 GA, 19 STRANDS OF NO. 34 WIRE WITH SILASTIC NO. 80 INSULATION

FIGURE 1 Outline drawing of electron tube type 8901
CONT.

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NOTES:-

1) DIMENSIONS IN INCHES. TOLERANCES ± .03 (.76)
2) MILLIMETERS IN PARENTHESES.
3) METRIC EQUIVALENTS (TO THE NEAREST .01) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM

FIGURE 2 Mounting fixture for temperature and temperature coefficient tests

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